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## Week 1 Workshop: Crafting Effective Questions and Hypotheses

### Scenario 1: Retail Sales Analysis

Context: You work for a retail company that has noticed a decline in sales during specific months. Your goal is to understand the factors contributing to this decline and formulate actionable insights to improve sales performance.

#### Tasks:

Craft an open-ended question that explores possible reasons for the sales decline.

- During months in which a significant decline in sales has been noted, what particular products are affected (as compared to a general reduction in sales storewide) and what are possible interactions with more static factors such as overall store demographics, target market, store/store location(s) if applicable, operations/logistics and more dynamic factors such as seasonality (eg. holidays, weather), changes in competitive landscape, demography shifts (expected vs. unexpected), as well as global/local context that could explain drops within a year or over multiple years?

Formulate a hypothesis that you will test using data analysis.

- If the data reveals recurrent, multi-month sales declines exceeding 15% below a 10-year average (or entire company history if less than 10 years old) — concentrated in specific products or product families rather than distributed evenly across all categories — then seasonal demand dynamics are the primary driver, because the affected products are likely those whose purchase timing is season-dependent and for which customers perceive competing options as more attractive at peak buying moments.

### Scenario 2: Healthcare Patient Satisfaction

Context: You are part of a healthcare organization aiming to enhance patient satisfaction. You want to identify areas where patient experiences could be improved.

#### Tasks:

Develop a closed-ended question to assess patient satisfaction with the current services.

- On a scale of 1 to 5 — where 1 is "very dissatisfied" and 5 is "very satisfied" — please rate your experience today in each of the following areas: (a) wait time from check-in to being seen by a provider, (b) clarity and thoroughness of the provider's explanation of your condition, diagnosis, and care plan, (c) courtesy and responsiveness of clinical and administrative staff, (d) coordination of your care including discharge instructions and follow-up planning, (e) cleanliness and comfort of the facility, and (f) your overall experience during this visit.
  - Question specifically for in-person outpatient visits that involve seeing a clinical provider
    - Given we are discussing a likely multi-faceted healthcare organization, question will need to be adapted for other settings such as imaging/lab visits (which would mean [a] would involve wait-time till service provided and [b] being removed), telehealth ([c] might need to have administrative staff removed if not used and [e] would not apply and might be better replaced by domain re: technical issues), and hospital stays ([a] might be better suited to be related to length of stay)

Frame a hypothesis that predicts the factors that most significantly impact patient satisfaction scores.

- If patient satisfaction scores, when disaggregated by domain, reveal that a singular domain correlates (correlation coefficient +/- 0.70) more strongly with overall satisfaction than others — and that this pattern holds consistently across multiple care settings within the organization — then structural and environmental factors such as facility cleanliness or technical infrastructure have comparatively less causal influence on patient satisfaction, because the cross-setting consistency of that dominant domain suggests that patient reflection on their experience is more heavily shaped by individual, visit-level interactions, such as interpersonal communication with providers and staff, than by fixed organizational or environmental conditions that would be expected to vary across settings.

### Scenario 3: Social Media Engagement

Context: A social media platform seeks to understand factors affecting user engagement with its app. Your objective is to uncover insights that can boost user interaction and retention.

#### Tasks:

Create an open-ended question that delves into user preferences and behaviors.

- Among platform superfans — defined as users in the highest tier of platform engagement by interaction frequency, content sharing, and community contribution — what specific content types drive the highest volume of interactions, and to what extent does engagement with those content types remain stable over time versus shifting in response to internal platform conditions such as new features or notification changes, or external factors such as demographic shifts or changes in the competitive landscape of other platforms?

Construct a hypothesis that links specific app features to higher user engagement.

- If users with the highest engagement levels (defined as the top quartile) are shown to achieve that engagement primarily through direct interactions with other users in the app — such as comments, replies, direct messages, and shared content exchanges — rather than through passive content consumption, then superfans will disproportionately be those with larger and more reciprocally active social networks on the platform, because repeated direct user-to-user contact both reflects an intrinsic reliance on the platform's social infrastructure and progressively raises the cost of switching to a competing platform where those specific relationships and interaction histories do not exist.

#### Scenario 4: Environmental Conservation

Context: An environmental organization aims to reduce plastic waste in a coastal area. You want to identify key locations where plastic waste accumulates the most.

#### Tasks:

Devise a closed-ended question to gather data on plastic waste accumulation in different areas.

- At each surveyed coastal site, using standardized transect sampling methodology (items per square meter of strandline), how does the count of plastic debris compare across sites as categorized by: (a) proximity to urban centers (within 10km, 10–50km, or >50km), (b) primary human activity type (residential, commercial fishing, tourism, industrial, or undisturbed), (c) visitor activity intensity (high: >500 visitors/day, moderate: 100–500 visitors/day, low: <100 visitors/day, or none/undisturbed), (d) waste management service frequency (daily, weekly, infrequent, or none), and (e) ocean current exposure (high, moderate, or low based on site orientation and regional current maps)?

Frame a hypothesis that predicts the factors contributing to higher plastic waste concentration.

- If higher plastic waste concentration is primarily driven by singular, non-repeat interactions between humans and the coastline, then the highest plastic waste item counts per square meter will be observed in tourism-categorized sites with high visitor activity intensity — with that effect exceeding the accumulation differences attributable to proximity to urban centers, waste management service frequency, or ocean current exposure — because tourists are, by definition, transient visitors with no ongoing affiliation to the coastline in question, are likely more prone to bringing single-use plastics such as bottles and packaging, and have comparatively little incentive to remove waste at the end of their visit relative to residents, commercial fishermen, and industrial workers who return repeatedly and have a demonstrable stake in the long-term condition of the site.

#### Approach Summary

Context: After completing the tasks for the above scenarios, create a brief summary answering the following questions.

#### Questions:

- How did you approach the task of crafting effective questions?
- What challenges did you face in formulating testable hypotheses?
- How might your questions and hypotheses evolve as you gather initial data?
- What additional information would have been helpful in refining your approach?
- How could your questions and hypotheses lead to actionable business insights?

For each scenario, I started by identifying the most likely organizational intent behind the task — for example, in Scenario 3, a social media platform would logically prioritize understanding its most engaged users before investigating churn-prone casual users, since superfans reveal what makes the platform uniquely valuable and what levers can be improved or marketed to attract new users. From there, I identified the most relevant domain-specific factors for each question. For Scenario 2, I searched peer-reviewed literature and found multiple research articles on patient satisfaction determinants, which directly informed the domains included in the closed-ended question.

Honestly, the vagueness of the scenarios was the main obstacle. With no actual data,

baseline metrics, or organizational context, keeping hypotheses narrow and specific was difficult — Scenario 1's hypothesis ended up needing to define what even constitutes a significant sales decline before getting to the actual prediction. The structure I settled on helped: IF [observable pattern in the data] THEN [measurable finding] BECAUSE [causal mechanism]. The hardest part was consistently the "because" clause — it needed to be logically consistent with the if and then while actually explaining the mechanism rather than just restating it. In Scenario 4, for example, the easy answer was "more humans = more waste," but that ignores that different types of human interaction with a coastline produce meaningfully different waste patterns.

Questions and hypotheses would evolve primarily in scope and focus. If investigation into seasonal sales decline revealed only one affected month rather than a full season, or that affected months didn't align with seasonal patterns across different store locations, the focus would shift away from seasonality toward more singular or non-cyclical events. Initial data also tends to surface new questions and reveal domain categories that weren't initially considered, so the expectation is that early findings would generate as many new hypotheses as they resolve.

Organizational context above all else — baseline metrics, prior analyses already conducted, known constraints, and the specific decisions the organization is anticipating needing to make. Without knowing where this analysis sits in the broader organizational landscape or what direction it's meant to support, it's hard to determine what's actually worth measuring versus what's academically interesting but operationally irrelevant.

Each hypothesis was structured to distinguish modifiable variables from contextual ones. In Scenario 1, seasonality and competitor activity provide context; promotional timing is the lever. In Scenario 2, wait times and facility conditions are addressable at an organizational level, but interpersonal communication quality — the strongest satisfaction driver per the literature — requires more targeted intervention through hiring criteria and provider training. In Scenario 3, platform design decisions should center on growing and protecting social network depth, since that's what creates switching costs. In Scenario 4, confirming tourism as the primary driver matters because it focuses intervention resources on the right coastline areas and opens the next set of questions about what specific tourist behaviors and infrastructure gaps are driving accumulation.